Clinic Management System

1. Project Overview

The **Clinic Management System** is a software engineering project designed to help clinics manage their daily operations, organize medical information, and improve the efficiency of their workflows.

The system covers several interconnected modules, including patient management, medical analyses, surgery management, room scheduling, and billing.

2. Team Members and Responsibilities

Team Member	Role	Responsibilities
Samir	Product Manager	Defines and prioritizes requirements, maintains the product backlog, and ensures the project meets user needs.
Ayyoub	Project Manager	Coordinates the team, monitors progress, manages deadlines, and organizes meetings.
Ahmed	Frontend Developer	Designs and implements the user interface.
Mohamed	Backend Developer	Implements business logic, database operations, and application functionality.
Amine	Tester & Documentation	Tests the system, identifies bugs, verifies requirements, and maintains project documentation.

All team members are expected to collaborate on design decisions, reviews, and problem-solving.

3. Process Model

Our team will follow an **Agile iterative and incremental process** because the Clinic Management System includes several interconnected modules, such as patient management, medical analyses, surgery management, room scheduling, and billing.

Developing the system in small increments will allow us to test each module early, detect integration problems, and adapt to changing requirements.

The team will organize development into iterations, prioritizing essential features first and progressively integrating additional functionality. Each iteration will include planning, implementation, testing, and a review of the results.

4. Definition of Done

A task is considered **Done** only when all the following conditions are satisfied:

1. The feature meets its specified acceptance criteria.
2. The code is committed to the appropriate branch and merged through an approved pull request.
3. The feature passes relevant functional tests, with no known critical bugs.
4. The feature is integrated successfully and does not break existing functionality.
5. Relevant documentation is updated when necessary.

5. Meetings and Communication

- **Weekly team meeting:** The team meets once a week to review project progress, discuss completed tasks, identify blockers, and plan upcoming work.
- **Urgent meetings:** The Project Manager (Ayyoub) can call an urgent meeting whenever an important issue arises or a change requiring team discussion is needed.
- **Task tracking:** GitHub Issues and GitHub Projects will be used to assign tasks, track progress, and document technical problems.
- **Team communication:** Members are expected to communicate blockers promptly and collaborate to resolve them.

6. Top Three Project Risks

Risk exposure is estimated using the following formula:

Risk Exposure = Probability × Impact

Probability and impact are scored from 1 (very low) to 5 (very high). These scores are initial team estimates and should be reviewed throughout development.

Risk 1: Sensitive Data Exposure

- **Probability:** 3/5 — Medium
- **Impact:** 5/5 — Very High
- **Risk Exposure:** 15/25

Description: The system handles sensitive medical and personal information. Unauthorized access or a data leak could harm patients, damage the clinic's reputation, and lead to legal or regulatory consequences.

Response:

- Prioritize security throughout development.
- Implement secure authentication and role-based access control.
- Validate inputs and protect against unauthorized access.
- Restrict access to sensitive information to authorized personnel.
- Avoid storing passwords in plain text and use secure password hashing.
- Test security controls and avoid using real patient data during development.

Risk 2: Incorrect or Incomplete Medical Information

- **Probability:** 4/5 — High
- **Impact:** 5/5 — Very High
- **Risk Exposure:** 20/25

Description: Missing, incorrect, or inconsistent medical information could cause operational delays, miscommunication, and potentially affect medical decisions.

Response:

- Make essential fields mandatory.
- Validate data formats and acceptable values.
- Clearly identify missing or incomplete information.
- Allow authorized personnel to correct errors and update records.
- Test validation rules and relationships between medical records.
- Ensure the system supports accurate data entry without replacing professional medical judgment.

Risk 3: Poor Data Migration

- **Probability:** 5/5 — Very High
- **Impact:** 4/5 — High
- **Risk Exposure:** 20/25

Description: Introducing the system into a clinic that currently relies on paper records may make transferring existing information difficult, time-consuming, and error-prone.

Response:

- Provide structured forms for entering historical patient information.
- Explore document scanning and OCR (Optical Character Recognition) to assist with digitization.
- Verify extracted information manually, especially critical medical details.
- Test the migration process using sample records before full deployment.
- Keep original paper records until the transferred information has been checked and validated.
- Plan the migration in stages to reduce disruption to clinic operations.

7. Risk Monitoring

The team will review these risks during weekly meetings. If a risk becomes more likely or its impact increases, the team will update its assessment and prioritize appropriate mitigation tasks.

8. Conclusion

This Team Process Charter establishes the team's responsibilities, development approach, quality standards, communication practices, and risk management

strategy. By following these guidelines, the team aims to deliver a reliable, secure, and maintainable Clinic Management System within the project deadline.